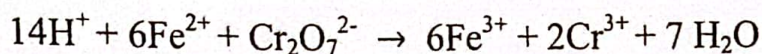


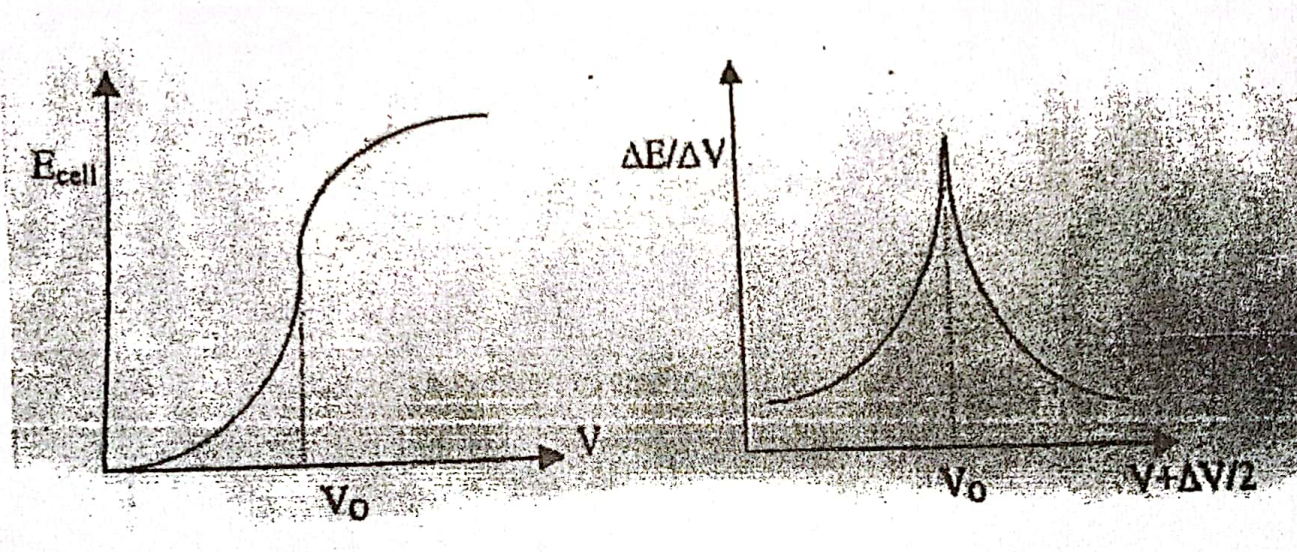
Experiment: Determination of concentration of Fe^{2+} Ions by titration with $\text{K}_2\text{Cr}_2\text{O}_7$ potentiometrically

Theory:

The redox reaction taking place during the reaction is given by,



The equivalence point (V_o) of the titration can be found from the following graph.



Procedure:

Pipette out 25.00 cm^3 of the Fe^{2+} solution into a titration flask, and dip a bright platinum wire electrode into it. Connect this solution to a calomel electrode via a narrow U-tube containing agar jelly alternatively. Titrate this Fe^{2+} solution with $0.01 \text{ mol cm}^{-3} \text{ K}_2\text{Cr}_2\text{O}_7$ in the burette by measuring the cell e.m.f. with a potentiometer. Take numerous readings in the neighborhood of the equivalence point where the e.m.f. changes rapidly and carry out the titration well beyond the equivalence point.

Results and calculations

1. Plot the e.m.f. values (E_{cell}) against volume of $\text{K}_2\text{Cr}_2\text{O}_7$ added (V) and deduce the equivalence point (V_o).
2. Plot $\Delta E / \Delta V$ against $V + \Delta V / 2$ and find the equivalence point (V_o) accurately.
3. Calculate the concentration of the Fe^{2+} solution.